

MNC Livestock Report

Technical Guide

Mobile boma movements, cattle counts, livestock predation, and illegal grazing reporting

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Workflow id: `mnc_livestock_report`

1. Overview

The **mnc_livestock_report** workflow fetches livestock-related events from EarthRanger for a specified time window — specifically **mobile_boma_rep**, **cattle_count**, **livestock_predation_rep**, and **illegal_grazing_rep** event types — and routes them into four independent reporting branches. In parallel, the workflow downloads MNC conservancy boundary and parcels geospatial files from Dropbox to use as base layers on all maps.

The workflow delivers:

- **mobile_boma_movement_summary_table.csv** — daily count of mobile boma movement events with a grand total row
- **total_cattle_count_summary_table.csv** — cattle counts per grazing zone per date
- **livestock_predation_summary_table.csv** — predation incidents by species, suspected predator, and number of animals affected
- **boma_movement_map.html / .png** — point map of mobile boma locations on MNC grazing zones and parcels
- **livestock_predation_events.html / .png** — point map of predation incidents on conservancy boundaries and parcels
- **illegal_grazing_map.html / .png** — point map of illegal grazing incidents on MNC grazing zones

Output summary

Output	Type	Source event type	Description
mobile_boma_movement_summary_table.csv	CSV	mobile_boma_rep	Daily boma event counts + Total row
total_cattle_count_summary_table.csv	CSV	cattle_count	Cattle per zone per date
livestock_predation_summary_table.csv	CSV	livestock_predation_rep	Predation records by species and predator
boma_movement_map.html / .png	Map	mobile_boma_rep	Boma locations on grazing zones and parcels
livestock_predation_events.html / .png	Map	livestock_predation_rep	Predation locations on conservancy boundaries
illegal_grazing_map.html / .png	Map	illegal_grazing_rep	Illegal grazing locations on grazing zones

2. Dependencies

2.1 Python packages

Package	Version	Channel
ecoscope-workflows-core	0.22.18.*	ecoscope-workflows
ecoscope-workflows-ext-ecoscope	0.22.18.*	ecoscope-workflows
ecoscope-workflows-ext-custom	0.0.43.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ste	0.0.18.*	ecoscope-workflows-custom
ecoscope-workflows-ext-mnc	0.0.8.*	ecoscope-workflows-custom

2.2 Connections and external assets

Asset	Task / Source	Purpose
EarthRanger	set_er_connection	Fetch event records and resolve event detail display titles (used in all process_events_details calls)
mnc_conservancy.gpkg	fetch_and_persist_file (Dropbox)	MNC community conservancy boundaries split by grazing zone. Used as polygon layers on the boma movement and illegal grazing maps.
mnc_across_the_river_parcel.gpkg	fetch_and_persist_file (Dropbox)	MNC across-the-river land parcels. Used as an additional polygon layer on the boma movement and livestock predation maps.

Note: Both Dropbox files are downloaded with `overwrite_existing: false` and `retries: 3`. If the files already exist in `ECOSCOPE_WORKFLOWS_RESULTS` from a previous run, the download is skipped.

2.3 Grouper

The workflow uses an **empty grouper list** (`groupers: []`). All event records are processed as a single undivided dataset — no fan-out or per-group branching is applied to the data. The grouper is passed through to the temporal index and the dashboard only.

3. Geospatial Asset Pipeline

Before any event data is fetched, the workflow downloads and prepares all geospatial base layers. These layers are shared across all three maps produced by the workflow.

3.1 Conservancy boundaries

Step	Task	Detail
1	fetch_and_persist_file	Download mnc_conservancy.gpkg from Dropbox to ECOSCOPE_WORKFLOWS_RESULTS (overwrite_existing: false, retries: 3).
2	load_df	Load the gpkg into a GeoDataFrame (layer: null, deserialize_json: false).
3	split_gdf_by_column	Split the GeoDataFrame into a dict keyed by the grazing_zone column values (Conservancy, Conservancy Herd Zone, Grazing Zone 1–4).
4	annotate_gdf_dict_with_geom_type	Add a geometry-type attribute to each GDF in the dict (required for layer rendering).

3.2 Styled zone layers

Two separate sets of DeckGL layers are created from the annotated dict:

Layer set	Task	Zones included	Used on maps
create_mnc_styled_layers	create_deckgl_layers_from_gdf_dict	Conservancy (grey outline), Conservancy Herd Zone (green), Grazing Zones 1–4 (dark olive, teal, dark green, sage)	Boma movement map, Illegal grazing map
create_conservancy_boundaries	create_deckgl_layers_from_gdf_dict	Conservancy boundary only (grey outline, no fill)	Livestock predation map

Note: The full zone style set includes a map legend with colour swatches for all six zone types. The conservancy-only set uses a single 'Boundaries' legend entry.

3.3 Conservancy and grazing zone GDFs

Step	Task	Detail
1	create_gdf_from_dict	Extract the Conservancy key from the split dict to produce a single-zone GeoDataFrame (conservancy_gdf). Used to place conservancy name text labels on maps.
2	filter_df	Filter the full loaded GDF to rows where grazing_zone != 'Conservancy' (op: ne). Produces overall_grazing_zones, used to compute the global map zoom and view state.

3.4 Conservancy text labels

Task: **create_custom_text_layer**. Renders conservancy names on the map using the **name** field from **conservancy_gdf**. Key style parameters:

Parameter	Value
get_text	name
get_color	[0, 0, 0, 255] (black)
get_size	1500 m
size_min_pixels	70
size_max_pixels	100
size_scale	2.25
font_family	Calibri
font_weight	700 (bold)
billboard	true
use_centroid	true (label placed at polygon centroid)

3.5 Parcels layer

Step	Task	Detail
1	fetch_and_persist_file	Download mnc_across_the_river_parcelsgpkg from Dropbox (overwrite_existing: false, retries: 3).
2	load_df	Load the parcels gpkg into a GeoDataFrame.
3	get_gdf_geom_type	Detect and attach the geometry type of the parcels GDF.
4	create_deckgl_layer_from_gdf	Render as a filled polygon layer: dark khaki fill (#bdb76b), opacity 0.15, stroked. Legend: 'Parcels'.

4. Event Ingestion Pipeline

All four reporting branches share a common event ingestion and normalisation pipeline.

4.1 Event retrieval

Parameter	Value
Task	get_events
event_types	mobile_boma_rep, cattle_count, livestock_predation_rep, illegal_grazing_rep
Columns retained	id, time, event_type, event_category, reported_by, serial_number, geometry, created_at, event_details, patrols
include_details	true
raise_on_empty	true
include_null_geometry	false
include_updates	false
include_related_events	false
include_display_values	false

4.2 Date extraction and temporal indexing

Step	Task	Detail
1	extract_column_as_type	Extract the time column as output_type: date into a new column named date .
2	add_temporal_index	Add a temporal index using time_col: date , groupers: [], cast_to_datetime: true, format: mixed. Produces the shared events_temporal DataFrame.

4.3 Common branch normalisation pattern

Every branch applies the same four-step normalisation after filtering:

Step	Task	Detail
1	filter_df	Filter events_temporal by event_type (op: equal, reset_index: true).
2	process_events_details	Resolve event detail field IDs to display titles (map_to_titles: true, ordered: true). Requires the ER client.
3	normalize_json_column	Flatten the event_details JSON column (skip_if_not_exists: true, sort_columns: true).
4	drop_column_prefix	Remove the event_details__ prefix from all flattened columns (duplicate_strategy: keep_original).

Note: Because `map_to_titles` is true, the flattened column names are the human-readable field titles from EarthRanger (e.g. 'Mobile Boma Zone', 'Livestock Species'). All downstream `map_columns` steps reference these titles directly.

4.4 Global map zoom value

Task: **`view_state_deck_gdf`**. Computes the map centre and zoom from the **`overall_grazing_zones`** GeoDataFrame (pitch: 0, bearing: 0). This view state is shared by the boma movement map and the illegal grazing map. The livestock predation map uses a fixed view state instead (lon: 35.2093, lat: -1.2578, zoom: 9.75).

5. Branch 1 — Mobile Boma Movements

Filters **mobile_boma_rep** events and produces a daily movement summary table and a point map of boma locations.

5.1 Normalisation

Steps 1–4 follow the common normalisation pattern in Section 4.3.

5.2 Column selection

Task: **map_columns** (raise_if_not_found: false, rename_columns: {}). The following columns are retained:

Column retained	Notes
id	Used for counting events in the summary
date	Used for grouping in the summary table
geometry	Used for the map
Date of Relocation	Event detail field (title)
Electric Boma Status	Event detail field (title)
Mobile Boma Zone	Event detail field (title)
Nature of the Site	Event detail field (title)
Reason for relocation	Event detail field (title)

5.3 Summary table

Step	Task	Detail
1	summarize_df	Group by date ; compute sum(id) displayed as boma_events (decimal_places: 0). reset_index: true.
2	add_totals_row	Append a grand Total row summing the boma_events column (label_col: date, label: 'Total').
3	persist_df	Save as mobile_boma_movement_summary_table.csv .

5.4 Map

Step	Task	Detail
1	exclude_geom_outliers	Remove spatial outliers using z_threshold: 3 on the map_mobile_boma GDF.
2	drop_null_geometry	Drop any remaining rows with null geometry.
3	apply_color_map	Colour points by event_type using the tab20 colormap → output column event_type_colors .

Step	Task	Detail
4	create_scatterplot_layer	Render points: get_radius: 4, opacity: 0.75, stroked: true. Legend title: 'Boma Movements', label from event_type, colour from event_type_colors.
5	combine_deckgl_map_layers	Static layers: create_mnc_styled_layers, create_mnc_parcel_layers, conservancy_text_layer. Grouped: mobile boma point layer.
6	draw_map	Render map (max_zoom: 10, legend placement: bottom-right, view_state from global_zoom_value).
7	persist_text	Save HTML to boma_movement_map.html .
8	html_to_png	Convert to PNG (device_scale_factor: 2.0, wait: 40 s).

6. Branch 2 — Cattle Counts

Filters **cattle_count** events and produces a table of cattle counts broken down by grazing zone per date. This branch produces no map.

6.1 Normalisation

Steps 1–4 follow the common normalisation pattern in Section 4.3.

6.2 Column selection and renaming

Task: **map_columns** (raise_if_not_found: false). The following columns are retained and renamed:

Source column (title after prefix drop)	Renamed to
date	date (retained, not renamed)
# cattle in Zone 1 mobile boma	zone_1
# cattle in Zone 2/3 mobile boma	zone_2_3
# cattle in Zone 4	zone_4
total_cattle_counted_from_all_zones	total_count

6.3 Persistence

Step	Task	Detail
1	persist_df	Save as total_cattle_count_summary_table.csv (filetype: csv).

7. Branch 3 — Livestock Predation

Filters **livestock_predation_rep** events and produces both a point map and a cleaned summary table.

7.1 Normalisation

Steps 1–4 follow the common normalisation pattern in Section 4.3.

7.2 Column selection for mapping

Task: **map_columns** (raise_if_not_found: false, rename_columns: {}). The following columns are retained for both the map and the summary table:

Column retained	Notes
id	Row identifier
date	Event date
geometry	Used for the map
Livestock Species	Event detail field (title)
Suspected Predator	Event detail field (title)
Total livestock affected	Event detail field (title)

7.3 Map

Step	Task	Detail
1	exclude_geom_outliers	Remove spatial outliers (z_threshold: 3).
2	drop_null_geometry	Drop rows with null geometry.
3	apply_color_map	Colour points by Livestock Species using the tab20 colormap → output column colors .
4	create_scatterplot_layer	Render points: get_radius: 4, opacity: 0.75, stroked: true. Legend title: 'Livestock Species', label from Livestock Species, colour from colors.
5	combine_deckgl_map_layers	Static layers: create_conservancy_boundaries, create_mnc_parcel_layers, conservancy_text_layer. Grouped: predation point layer.
6	draw_map	Render map using fixed view state (lon: 35.2093, lat: -1.2578, zoom: 9.75, max_zoom: 10, legend: bottom-right).
7	persist_text	Save HTML to livestock_predation_events.html .
8	html_to_png	Convert to PNG (device_scale_factor: 2.0, wait: 40 s).

Note: The livestock predation map uses a fixed view state rather than the computed global_zoom_value, centred on the MNC area at zoom 9.75. It also uses the conservancy-boundaries-only layer set (no coloured grazing zones) to keep the focus on predation incident locations.

7.4 Summary tables

The livestock predation branch produces two independent CSVs from the same normalised DataFrame.

7.4a total_livestock_predation_summary_table.csv

Step	Task	Detail
1	summarize_df	Group by date ; compute nunique(id) displayed as livestock_predation_events (decimal_places: 0). reset_index: true.
2	add_totals_row	Append a grand Total row summing the livestock_predation_events column (label_col: date, label: 'Total').
3	persist_df	Save as total_livestock_predation_summary_table.csv .

7.4b livestock_predation_summary_table.csv

Step	Task	Detail
1	map_columns	Retain date, Livestock Species, Suspected Predator, Total livestock affected. Rename: Livestock Species → livestock_species, Suspected Predator → suspected_predator, Total livestock affected → total_livestock_affected. (raise_if_not_found: false)
2	replace_missing_with_label	Replace nulls in suspected_predator and livestock_species with the label 'Unknown' .
3	map_column_values	Map 'Other (specify in comments)' → 'Unknown' in the suspected_predator column (inplace: true).
4	convert_to_int	Cast total_livestock_affected to integer (errors: coerce, fill_value: 0, inplace: false).
5	persist_df	Save as livestock_predation_summary_table.csv .

8. Branch 4 — Illegal Grazing

Filters **illegal_grazing_rep** events and produces a point map of illegal grazing incidents on the MNC grazing zones. This branch produces no CSV summary table.

8.1 Normalisation

Steps 1–4 follow the common normalisation pattern in Section 4.3.

8.2 Column selection

Task: **map_columns** (raise_if_not_found: false, rename_columns: {}). The following columns are retained:

Column retained	Notes
date	Event date
event_type	Event type identifier; used for colouring map points
geometry	Used for the map
Herd Zone	Event detail field (title)
Landowner name	Event detail field (title)
action taken	Event detail field (title)

8.3 Map

Step	Task	Detail
1	exclude_geom_outliers	Remove spatial outliers (z_threshold: 3).
2	drop_null_geometry	Drop rows with null geometry.
3	apply_color_map	Colour points by event_type using the tab20 colormap → output column event_type_colors .
4	create_scatterplot_layer	Render points: get_radius: 4, opacity: 0.75, stroked: true. Legend title: 'Illegal grazing', label from event_type.
5	combine_deckgl_map_layers	Static layers: create_mnc_styled_layers, conservancy_text_layer. Grouped: illegal grazing point layer. (Note: parcels layer is not included on this map.)
6	draw_map	Render map (max_zoom: 10, legend: bottom-right, view_state from global_zoom_value).
7	persist_text	Save HTML to illegal_grazing_map.html .
8	html_to_png	Convert to PNG (device_scale_factor: 2.0, wait: 40 s).

9. Output Files

All outputs are written to **ECOSCOPE_WORKFLOWS_RESULTS**.

9.1 CSV tables

File	Branch	Columns	Description
mobile_boma_movement_summary_table.csv	Mobile Boma	date, boma_events	Daily boma event counts with a grand Total row
total_cattle_count_summary_table.csv	Cattle Count	date, zone_1, zone_2_3, zone_4, total_count	Cattle counts per zone per date
total_livestock_predation_summary_table.csv	Livestock Predation	date, livestock_predation_events	Daily unique predation event count with a grand Total row
livestock_predation_summary_table.csv	Livestock Predation	date, livestock_species, suspected_predator, total_livestock_affected	Predation records by species and predator

9.2 Maps

File	Branch	Coloured by	Base layers
boma_movement_map.html / .png	Mobile Boma	event_type	MNC grazing zones, parcels, conservancy labels
livestock_predation_events.html / .png	Livestock Predation	Livestock Species	Conservancy boundaries, parcels, conservancy labels
illegal_grazing_map.html / .png	Illegal Grazing	event_type	MNC grazing zones, conservancy labels (no parcels)

10. Workflow Execution Logic

10.1 Per-task skip conditions

This workflow does **not** use a global **task-instance-defaults** block. Every task from event retrieval onwards carries its own explicit skipif block:

Condition	Behaviour
any_is_empty_df	Skip this task if any input DataFrame is empty
any_dependency_skipped	Skip this task if any upstream dependency was skipped

Note: Because skip conditions are per-task rather than global, each of the four branches propagates skips independently. For example, if no cattle_count events are returned, only the cattle count branch is skipped; the other three branches continue normally.

10.2 Four independent branches

After the shared ingestion pipeline produces **events_temporal**, the workflow splits into four fully independent branches. Each branch reads directly from events_temporal with no cross-branch dependencies:

Branch	Filter value	CSV output	Map output
Mobile Boma	mobile_boma_rep	mobile_boma_movement_summary_table.csv	boma_movement_map
Cattle Count	cattle_count	total_cattle_count_summary_table.csv	—
Livestock Predation	livestock_predation_rep	livestock_predation_summary_table.csv	livestock_predation_events
Illegal Grazing	illegal_grazing_rep	—	illegal_grazing_map

10.3 No mapvalues or fan-out

This workflow processes all records as a single batch. There is no **mapvalues**, **split_groups**, or **zip_groupbykey** — every task runs exactly once.

10.4 HTML to PNG conversion

Three maps are rendered as HTML and then converted to PNG using **html_to_png**. All three use the same conversion settings: device_scale_factor: 2.0, wait_for_timeout: 40 000 ms, max_concurrent_pages: 1, full_page: false.

10.5 Dashboard

The workflow concludes with **gather_dashboard** which packages workflow details, time range, and groupers. The **widgets** list is empty — no single-value or map widgets are configured.

11. Software Versions

Package	Version pinned in spec.yaml
ecoscope-workflows-core	0.22.18.*
ecoscope-workflows-ext-ecoscope	0.22.18.*
ecoscope-workflows-ext-custom	0.0.43.*
ecoscope-workflows-ext-ste	0.0.18.*
ecoscope-workflows-ext-mnc	0.0.8.*